

Question:

Rob, Royce, and Pedro are contractors getting ready to put a new roof on three homes. If the three homes will need 250 cases of shingles, with the first house needing $\frac{1}{2}$ of the second, and the third needing double the first. How many cases of shingles will the third house need?

Reasoning + Answer:

`<think>` Let x be cases in 1st house, then $\frac{1}{2}x$ for 2nd and $2x$ for 3rd. Sum of all cases
= 250 = $x + (\frac{1}{2})x + 2x = 3.5x$. So $3.5x = 250$, thus $x = 250 / 3.5 = 71.43$ cases for
1st house. Then 2nd house needs $\frac{1}{2} * 71.43 = 35.71$ cases and 3rd needs $2 * 71$
 $.43 = 142.86$ cases. `</think>` `<answer>` 142.86 `</answer>`

